Topic 2: Cells, Viruses and Reproduction of Living Things

This topic considers the ultrastructure of prokaryotes, eukaryotes and viruses. Details of the types of nuclear division are included and how these are involved in animal and plant reproduction. Microscopy and observational skills are developed through the preparation of stained plant tissue.

Opportunities for developing mathematical skills within this topic include: recognising and using expressions in decimal and standard form; making order of magnitude calculations; changing the subject of an equation; plotting two variables from experimental or other data. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Students should:	
2.1	**Eukaryotic and prokaryotic cell structure and function**
i	Understand that cell theory is a unifying concept that states that cells are a fundamental unit of structure, function and organisation in all living organisms.
ii	Understand that in complex organisms, cells are organised into tissues, organs, and organ systems.
iii	Know the ultrastructure of prokaryotic cells and the structure of organelles, including: nucleoid, plasmids, 70S ribosomes and cell wall.
iv	Be able to distinguish between Gram positive and Gram negative bacterial cell walls and understand why each type reacts differently to some antibiotics.
v	Know the ultrastructure of eukaryotic cells and the functions of organelles, including: nucleus, nucleolus, 80S ribosomes, rough and smooth endoplasmic reticulum, mitochondria, centrioles, lysosomes, Golgi apparatus, cell wall, chloroplasts, vacuole and tonoplast.
vi	Know how magnification and resolution can be achieved using light and electron microscopy.
vii	Understand the importance of staining specimens in microscopy.
CORE PRACTICAL 2: Use of the light microscope, including simple stage and eyepiece micrometers and drawing small numbers of cells from a specialised tissue.	